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Antibacterial activity of leaf extracts of *Euphorbia hirta* L. against *Escherichia coli* and *Staphylococcus aureus*

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Abstract : The antibacterial activity of *Euphorbia hirta* L. were examined using agar disc diffusion method against *Escherichia coli* and *Staphylococcus aureus* . Leaves were extracted using different solvents such as methanol, ethanol, Dimethyl sulfoxide and aqueous. Among treatments, maximum *in vitro* inhibition was scored in methanol extracts of *Euphorbia hirta* L. which offered inhibition zone of 25 mm and 20 mm against *E. coli and S. aureus* respectively, followed by Dimethyl sulfoxide extract of the same plant leaf with inhibition zone of 20 mm and 15 mm respectively. A significant inhibition of *E. coli* was found in aqueous and ethanol solvent extracts of *Euphorbia hirta*. In case of *S. aureus*, significant inhibition of 15 mm was obtained in ethanol extracts and 13 mm from aqueous extract of *Euphorbia hirta* .

Keywords: Antibacterial activity, solvents, leaf extracts, *Escherichia coli* and *Staphylococcus aureus*.

INTRODUCTION

Plants produce a diverse range of bioactive molecules, making them a rich source of different types of medicines (Farombi, 2003).[1] Despite the existence of potent antibacterial agents, resistant or multi-resistant strains are continuously emerging, imposing the need for a continuous search and development of new drugs (Barbour et al., 2004; Machado et al., 2003; Rojas et al., 2003).[2-4] Hence, many efforts have been exploited to discover new antibacterial compounds from various kinds of sources such as soil, microorganisms, animals and plants. According to World Health Organization (Santos et al., 1995)[5] medicinal plants would be the best source to obtain a variety of drugs. In the present investigation, the potential of *Euphorbia hirta* leaf extracts for antibacterial activity against important human pathogenic bacteria has been evaluated.

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Euphorbia hirta has been widely used by Tribal as traditional medicine in a treatment against infectious pathogens. *Euhorbia hirta* Linn. is a perennial herb belonging to the family Euphorbiaceae. It is a common weed referred to as 'garden spurge'. It is a slender-stemmed, annual hairy plant with many branches. Leaves are opposite, elliptic – oblong to oblong – lanceolate, acute or subacute. It is a potent medicinal plant and has established its sedative and anxiolytic activity , analgesic, antipyretic, anti-inflammatory, antidepressant for blood pressure ,antihypertensive and antioxidant. It is now more important in treating respiratory ailments, especially cough, coryza, bronchitis and asthma .There is evidence to show that it has been used as a medicinal herb for various ailments like diarrhea, dysentery, warts, worm infestations etc .

Antibacterial activity of crude extracts of *Euphorbia hirta* against some few bacteria associated with enteric infections was studied by some scientist.((El-Mahmood et al.,2009;Ibrahim et al.,2012; Shanmugapriya et al.,2012).[6-8] Other workers have also shown that extracts

of *Euphorbia hirta* inhibited the growth of various microorganisms (Sunil Kumar et al.,2010; Suresh et al.,2008).[9,10]

MATERIALS AND METHODS

Plant material

Fresh leaves of *Euphorbia hirta* were collected from the campus of Nirmala College, Doranda and also from different locations of Ranchi district of Jharkhand ,India during the month of Nov. 2013- Feb 2014. Fresh leaves were initially separated and washed thoroughly 2 - 3 times with running tap water and then with sterile water followed by shade-dried, powdered and used for extraction.

Preparation of the Extract

Collected leaves were cleaned with water and dried in shade and pulverized into fine powdered substances by a grinding machine. Each 15 g of powder was transferred into conical flasks. Then 150 mL of methanol was added in the flasks, closed by foil paper and placed on a shaker at 37 °C temperature for 72 hr. The crude extracts were then filtered by passing the extracts through Whatman No. 1 filter paper and then concentrated. After complete solvent evaporation, extract was weighed and stored in a refrigerator at 4 °C for further use . 500 mg of solvent residue was dissolved in 10 mL of solvents were used as the test extracts for antibacterial activity assay.

Test Bacteria

Human pathogenic bacteria such as *Escherichia coli* and *Staphylococcus aureus* were collected from Birsa Agriculture University, Kanke , Ranchi, Jharkhand, India. All the test bacterial species were maintained on nutrient agar media.

Antibacterial Activity

Antibacterial activity of aqueous extract and solvent extracts; methanol, ethanol and Dimethyl sulfoxide were determined by disc diffusion method on nutrient agar medium . The filter paper discs of 5 mm diameter were prepared using Whatman No. 1 filter paper , soaked in extract . The discs dipped in respective solvent were used as negative controls. The petri-dishes were sterilized in hot air oven and nutrient agar medium was sterilized by autoclaving. This media was poured in the sterile petri-dishes . After 24 h ,inoculum containing bacteria were spread on the solid plates with a sterile swab moistened with the bacterial suspension. The impregnated discs were aseptically placed on the solidified agar media. After 24 h of incubation at 37 °C temperature the culture plates were examined and the diameters of the inhibition zones were measured in mm unit.

RESULTS AND DISCUSSION

The antibacterial activity of various solvent extracts of leaves of *E.hirta* against both gram - positive and gram-negative human pathogenic bacteria showed varied level of inhibition (Table -1 and Table - 2). Among treatments, maximum *in vitro* inhibition of tested bacteria *E. coli, S. aureus,* was scored in methanol extracts of *E.hirta* which offered Zone of inhibition of 25 and 20 mm respectively and Zone of Inhibition Area of 686.88 and 471.00 mm^2 respectively. Further, Dimethyl sulfoxide extract of leaves of *E.hirta* was effective against all two tested bacteria which recorded significant Zone of inhibition 20 and 15 mm respectively and Zone of Inhibition Area of 471.00 and 294.38 mm^2 respectively. A significant inhibition zone of clinical bacteria *E. coli* was found in aqueous and ethanol extracts of *E.hirta* showing 15 and 15 mm inhibition respectively. In case of human pathogenic bacteria *S. aureus,* significant Zone of inhibition of 15 mm and Zone of inhibition area of 294.38 mm^2 was obtained in ethanol extracts and 13 mm and 234.72 mm^2 respectively from aqueous extract of *E.hirta* .

From Table 1 and 2, it is seen that the leaf powder extract from methanol ,ethanol, Dimethyl sulfoxide and aqueous of *E. hirta* showed antibacterial activity against *E. coli* and *S. aureus* .Methanol extract of *E. hirta* leaf showed the maximum degree of antibacterial activity properties. The crude methanol extract of *E. hirta* leaf powder produced the highest 25 mm zone of inhibition against *E.coli* and 20 mm zone of inhibition against *S. aureus* .

This may be due to the presence of alkaloids, tannins, saponins and flavonoids which are plant secondary

Table 1: Study of Diameter of Zone of Inhibition (DIZ) and Zone of Inhibition Area (ZIA) of Extract of *E.hirta* leaf in Different Solvent against *E .coli*.

Solvent	Diameter of Disc (mm)	Diameter of Inhibition including disc (mm)	DIZ (mm)	ZIA(mm^2)
Methanol	5	30	25	686.88
Ethanol	5	20	15	294.38
Dimethyl sulfoxide	5	25	20	471.00
Aqueous	5	20	15	294.38

Table 2: Study of Diameter of Zone of Inhibition (DIZ) and Zone of Inhibition Area (ZIA) of Extract of *E.hirta* leaf in Different Solvent against *Staphylococcus aureus* .

Solvent	Diameter of Disc (mm)	Diameter of Inhibition including disc (mm)	DIZ (mm)	ZIA(mm^2)
Methanol	5	25	20	471.00
Ethanol	5	20	15	294.38
Dimethyl sulfoxide	5	20	15	294.38
Aqueous	5	18	13	234.72

metabolites known to possess antibacterial properties.
DIZ = Diameter of zone of inhibition in millimeter scale.
ZIA = Zone of Inhibition Area in millimeter square.

CONCLUSIONS

From the observed result of this project work it can be concluded that the *E. hirta* leaf extracts were found to be effective as a source of antibacterial agents against pathogenic bacteria and it was most effective against *E.coli* and *S. aureus* . These primary extracts open the possibility of finding new clinically effective antibacterial compounds. *E.hirta* providing active extracts are found in different locations of Ranchi .It is used for various medical purposes .Continued further exploration of plant-derived antimicrobials is needed today. Further research is necessary to determine the identity of the antibacterial compounds from within this plant and also to determine their full spectrum of efficacy. However the present study of in vitro antibacterial evaluation of *E.hirta* forms a primary platform for further phytochemical and pharmacological studies to discover new antibiotic drugs.

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